

§6.AK Gate-Wiring Integration Patch — exact, copy-pasteable

Date: 2026-06-26 **Status:** Ready to apply (logic proven hermetically — see §6 below) **Closes:** §6.AK-debt (the mechanical wiring of `scripts/check-cycle-coverage.sh` into the distribute gate + pre-push hook).

Companion test: `tests/cycle-coverage/test_wiring.sh` (5/5 PASS against the real gate; mutation-rehearsal proven). **Spec parent:** `docs/superpowers/specs/2026-06-26-ak-cycle-coverage-spec.md` §5.3 + §5.4.

Classification: project-specific (the two insertion patches are Lava-file-specific; the `gate-coverage-must-intersect-cycle-claims` pattern is universal per §6.AK).

IMPORTANT — this document does NOT edit the live gate files. It is the precise patch the main stream applies later. The two target files (`scripts/firebase-distribute.sh`, `.github/hooks/pre-push`) are pushed-through-pre-push concurrently by the main stream; editing them here would corrupt an in-flight gate. Apply these patches only when the main stream is quiescent.

0. Why these two insertion points

`scripts/check-cycle-coverage.sh` already exists and passes its own hermetic test (`tests/cycle-coverage/test_cycle_coverage.sh`, 7/7).

§6.AK-debt names exactly two unwired call-sites:

1. **`scripts/firebase-distribute.sh`** — a Phase-1 Gate that REFUSES the distribute at run time when the cycle-coverage gate fails (spec §5.3).
2. **`.github/hooks/pre-push`** — a Check that REJECTS a push which *advances a distribute last-version pointer* while the cycle-coverage gate fails (spec §5.4).

Both call the SAME committed gate with the SAME evidence file; they compose, they do not duplicate.

1. Patch A — scripts/firebase-distribute.sh (Phase-1 Gate 7)

1.1 Exact insertion point

Insert the new block **immediately after the Phase-1 Gates 4+5 block closes** and **before the # 2. Resolve git SHA section comment**. The anchor in the current file:

```
304 else
305     echo "    Phase 1 Gates 4+5 skipped: .env not present
    (auth feature inert at runtime; runtime falls back to
    StubLavaAuthBlobProvider)."
306 fi
307                                     ← INSERT THE NEW BLOCK ON THIS BLANK
    LINE (between line 306 `fi` and line 308 comment)
308 # -----
309 # 2. Resolve git SHA + branch for the release notes
```

All variables the block references are already in scope at line 306: `MODE` (parsed @46–61), `SCRIPT_DIR` (@32), `APP_VERSION / APP_VERSION_CODE` (@113–116), `CHANGELOG_DIR` (@140, app-resolved). The block uses `NO` variable defined after line 306.

1.2 Block to paste (between line 306 and line 308)

```
# -----
# 1c. §6.AK Phase-1 Gate 7 – cycle-coverage (claims × executed
#     device Challenges).
# Closes the firebase-distribute portion of §6.AK-debt. Refuses to
#     distribute
# unless EVERY CHANGELOG-claimed user-visible fix for this version
#     has an
# EXECUTED+PASSED covering device Challenge in the §6.Z evidence
#     file for the
# SAME commit SHA (spec §5.3). This is the gate that would have
#     CAUGHT the 1076
# incident (commit 627a0d58): a C00-only device gate while the
#     CHANGELOG claimed
# search / provider-selection / onboarding fixes.
#
# NOTE: the gate's default --evidence-dir for the debug channel
#     resolves to
# .../firebase-app-distribution-dev, but the client+api-app
#     distribute layout
# keeps the §6.Z evidence in $CHANGELOG_DIR (firebase-app-
#     distribution for
# client, -api-app for api-app). We therefore pass --evidence-
#     dir="$CHANGELOG_DIR"
```

```

# explicitly. The cycle-coverage-map auto-resolves to
# $CHANGELOG_DIR/${APP_VERSION}-${APP_VERSION_CODE}-cycle-
# coverage-map.yaml
# (the cycle author writes it alongside the $6.Z evidence).
#
# This block also subsumes the still-OPEN $6.Z runtime gate ($6.Z-
# debt): the
# cycle-coverage gate itself asserts evidence-presence + commit-
# SHA match +
# ≤24h freshness (exit 2 / exit 1) before checking claim coverage.
# -----
case "$MODE" in
    release) AK_CHANNEL="release" ;;
    *)       AK_CHANNEL="debug"   ;;
esac
echo "    Phase 1 Gate 7 ($6.AK): cycle-coverage – CHANGELOG
      claims × executed device Challenges"
ak_rc=0
"$SCRIPT_DIR/check-cycle-coverage.sh" \
    --version="$APP_VERSION-$APP_VERSION_CODE" \
    --channel="$AK_CHANNEL" \
    --evidence-dir="$CHANGELOG_DIR" \
    --strict || ak_rc=$?
case "$ak_rc" in
    0) echo
        "    $6.AK gate PASS – all CHANGELOG claims covered by
        executed device Challenges" ;;
    1) echo "FATAL $6.AK: CHANGELOG claim(s) lack a covering
        executed+PASSED device Challenge for $APP_VERSION-
        $APP_VERSION_CODE." >&2
        echo "    Run the missing device Challenge(s) on the
        gate, OR strike the unverified claim(s) from CHANGELOG.md
        ($6.AK clause 6)." >&2
        exit 1 ;;
    2) echo "FATAL $6.AK: $6.Z evidence or cycle-coverage-map
        missing / stale / wrong-SHA for $APP_VERSION-
        $APP_VERSION_CODE." >&2
        echo "    Re-run the device gate on THIS commit and
        write the cycle-coverage-map before distributing." >&2
        exit 1 ;;
    *) echo
        "FATAL $6.AK: internal error in cycle-coverage scanner
        (rc=$ak_rc)." >&2
        exit 1 ;;
esac

```

1.3 set -e correctness note (load-bearing)

firebase-distribute.sh runs under `set -euo pipefail (@30)`. The gate exits non-zero on a covered-claim failure; the `|| ak_rc=$?` idiom is what prevents `set -e` from aborting the script *before* the case can emit the \$6.AK-specific FATAL message. Do **not** rewrite this as a bare

"\$SCRIPT_DIR/check-cycle-coverage.sh" ...; case \$? in — under set -e the bare call would abort on a non-zero exit and the operator would never see the §6.AK directive. (This is exactly the idiom the wiring test proves in case `firebase_coverage_fail`.)

1.4 Scope: applies to BOTH apps

The block uses `$CHANGELOG_DIR` (app-resolved in `$0` of the script: `firebase-app-distribution` for `--app client`, `firebase-app-distribution-api-app` for `--app api-app`). It therefore gates **both** artifacts per §6.AK clause 7. Implication for the main stream: **api-app distributes now also require** a `${APP_VERSION}-${APP_VERSION_CODE}-test-evidence.{md,json} + ${APP_VERSION}-${APP_VERSION_CODE}-cycle-coverage-map.yaml` under the `api-app` channel dir. If the first post-graft `api-app` distribute has no claims, ship a map with the mandatory C00/C01 minimum per spec §3.3 (or an empty-CHANGELOG cycle records zero feature claims — but the map **MUST** still exist, else exit 2). If the main stream wants to defer `api-app`, wrap the block in `if [[ "$SELECTED_APP" == "client" ]]; then ... fi` and record the `api-app` deferral as an explicit §6.AK-debt sub-item — but the constitutionally-correct default is universal.

2. Patch B — .githooks/pre-push (Check 10)

2.1 Exact insertion point

Insert the new Check **immediately after Check 7's for ptr ... done loop closes** and **before the # ===== Check 9 comment**. The anchor in the current file:

```
191         fi
192     fi
193     done                                     ← this `done` closes Check
        7's `for ptr in last-version-debug last-version-release`
194                                     ← INSERT THE NEW CHECK 10
        BLOCK ON THIS BLANK LINE (between 193 and 195)
195     # ===== Check 9: §11.4.18 / CM-SCRIPT-DOCS-SYNC – script
        doc sync =====
```

The block lives inside the per-SHA for `sha in $range; do ... done` loop (which spans lines 58–261) and appends to the `violations` array exactly like every other Check. It reuses Check 7's proven advance-detection idiom (`new_val > old_val` via `git show "$sha^:..."`).

2.2 Block to paste (between line 193 and line 195)

```
# ===== Check 10: $6.AK cycle-coverage on distribute-
# advancing commits =====
# Closes the pre-push portion of $6.AK-debt (companion to
# firebase-
# distribute.sh's Phase-1 Gate 7). When a commit ACTUALLY
# advances

# last-version-debug or last-version-release (NEW > OLD -
# same detection as

# Check 7), a pointer-advance signifies "ready to
# distribute"; the $6.AK
# cycle-coverage gate MUST then pass for that version: every
# CHANGELOG-claimed
# user-visible fix needs an executed+PASSED covering device
# Challenge in the
# $6.Z evidence file. Pushing the advance without covering
# device evidence is
# the 1076 incident shape (627a0d58: C00-only gate shipped
# broken flows).
#
# NOTE: this mirrors Check 7's CLIENT-channel scope (firebase-
# app-distribution).
# api-app pointer advances live under firebase-app-
# distribution-api-app and
# are gated at distribute time by firebase-distribute.sh's
# Gate 7; extending
# Check 10 to the api-app channel is an owed follow-up (track
# under $6.AK-debt).
ak_chan_dir=".lava-ci-evidence/distribute-changelog/firebase-
app-distribution"
for ptr in last-version-debug last-version-release; do
    ak_ptr_path="${ak_chan_dir}/${ptr}"
    git diff-tree --no-commit-id --name-only -r "$sha" 2>/dev/
    null | grep -qF "$ak_ptr_path" || continue
    ak_old=$(git show "$sha^:$ak_ptr_path" 2>/dev/null | tr -d
    ' \n' || echo "0")
    ak_new=$(git show "$sha:$ak_ptr_path" 2>/dev/null | tr -d '
    \n' || echo "0")
    [[ "$ak_new" =~ ^[0-9]+$ && "$ak_old" =~ ^[0-9]+$ &&
    "$ak_new" -gt "$ak_old" ]] || continue
    case "$ptr" in
        last-version-debug)    ak_channel="debug" ;;
        last-version-release)  ak_channel="release" ;;
        *)                     ak_channel="debug" ;;
    esac

    # Resolve versionName for this advance: per-version
    # snapshot first
    # (firebase-distribute writes pointer + snapshot
    # atomically), then fall
```

```

# back to build.gradle.kts versionName at this commit.
ak_vname=""
ak_snapshot=$(ls "${ak_chan_dir}"/*-${ak_new}.md 2>/dev/
null | grep -v '\-test-evidence\.md$' | head -1 || true)
if [[ -n "$ak_snapshot" ]]; then
    ak_vname=$(basename "$ak_snapshot" -${ak_new}.md)
else
    ak_vname=$(git show "$sha:app/build.gradle.kts" 2>/dev/
null | grep -oE 'versionName[[:space:]]*=[[:space:]]*"^[^
+]"' | head -1 | sed 's/\..*\[^\]*\)\.*/\1/' || echo "")
fi
[[ -n "$ak_vname" ]] || continue
if [[ -x ./scripts/check-cycle-coverage.sh ]]; then
    ak_rc=0
    ./scripts/check-cycle-coverage.sh \
        --version="${ak_vname}-${ak_new}" \
        --channel="$ak_channel" \
        --evidence-dir="$ak_chan_dir" \
        --head="$sha" \
        --strict >/dev/null 2>&1 || ak_rc=$?
    if [[ "$ak_rc" -ne 0 ]]; then
        violations+=("$sha: §6.AK violation – advances $ptr $
{ak_old}→${ak_new} (${ak_vname}) but the cycle-coverage
gate exits $ak_rc. A CHANGELOG claim lacks a covering
executed+PASSED device Challenge (or the §6.Z evidence /
cycle-coverage-map is missing / stale / wrong-SHA). Run
the missing device Challenge(s) on the gate, OR strike the
unverified claim(s) from CHANGELOG.md (§6.AK clause 6),
before pushing the pointer advance.")
    fi
fi
done

```

2.3 SHA-tie caveat (must be understood before applying)

Check 10 passes `--head="$sha"` (the advancing commit). `check-cycle-coverage.sh` therefore requires the §6.Z evidence file's `cycle-coverage:header commit=` to equal that advancing commit (else exit 2 → violation). This matches `firebase-distribute.sh`'s existing **atomic** `pointer+snapshot write` (@484–512): the cycle author records the §6.Z evidence (with the build commit) in the **same** commit that advances the pointer. If a future workflow advances the pointer in a *separate* later commit from the one that wrote the evidence, the evidence `commit=` must be amended to that pointer-advancing commit, or Check 10 will (correctly, per §6.AK) reject the push. The `violations[]` message names this so the operator can act.

2.4 §6.N.1.2 + Check 9 obligations on the GRAFT COMMIT (do not skip)

The commit that applies BOTH patches will itself trip the existing hook:

- **Check 4 / §6.N.1.2** fires because the diff touches gate-shaping files (`.githubhooks/pre-push` AND `scripts/firebase-distribute.sh`, both in Check 4's list @97–98). The graft commit body **MUST** carry a Bluff-Audit: stamp that NAMES at least one file in the diff. Cite `tests/cycle-coverage/test_wiring.sh` as the rehearsal. Example stamp:

Bluff-Audit: `.githubhooks/pre-push + scripts/firebase-distribute.sh` (§6.AK Check 10 + Gate 7)
Mutation: swallow the §6.AK refusal (`firebase `exit 1` → no-op; pre-push `violations+=` → no-op`)
Observed-Failure: `tests/cycle-coverage/test_wiring.sh` reports
 "FAIL [firebase_coverage_fail] exit=0 (expected 1)" and
 "FAIL [prepush_advance_fail] exit=0 (expected 1)" → RESULT:
FAIL (exit 1);
 the firebase stub even prints "distribute would proceed" after a failing gate.
Reverted: yes
 - **Check 9 / CM-SCRIPT-DOCS-SYNC** fires because `docs/scripts/firebase-distribute.sh.md` already EXISTS — so the graft commit **MUST** also update that doc in the SAME commit (add a short "Phase 1 Gate 7 (§6.AK)" subsection). `.githubhooks/pre-push` is not under `scripts/*.sh`, so Check 9 does NOT apply to it.
 - **Owed gap (flag, not blocking):** `docs/scripts/check-cycle-coverage.sh.md` is ABSENT. Check 9 only fires when `check-cycle-coverage.sh` is *modified* (we don't modify it here), so it does not block this graft — but the companion doc for that script is owed under CM-SCRIPT-DOCS-SYNC and should be authored next.
-

3. Evidence-file + map contract the cycle author must satisfy

For the Gate-7 / Check-10 pair to PASS on a real distribute, the cycle author writes two files under the channel dir (`.lava-ci-evidence/distribute-changelog/firebase-app-distribution/` for client):

1. `<vname>-<code>-test-evidence.md` (the §6.Z device-gate evidence) containing:
 - exactly one header line: `cycle-coverage: version=<vname>-<code> commit=<sha> channel=<debug|release> timestamp=<ISO-8601 ≤24h>`
 - one challenge: line per executed Challenge: `challenge: fq=<FQN> verdict=<PASS|FAIL|SKIP> runner=<containers-submodule|genymotion-vm|...>` (a covering Challenge counts only when `verdict=PASS AND runner != host-direct`, per §6.AH).
2. `<vname>-<code>-cycle-coverage-map.yaml` mapping each CHANGELOG user-visible bullet to its covering_challenge (spec §3.1).

These are the same two file kinds the sibling gate test (`tests/cycle-coverage/test_cycle_coverage.sh`) and the wiring test (`tests/cycle-coverage/test_wiring.sh`) synthesize.

4. Apply order (recommended)

1. Verify the main stream is NOT mid-push (the two target files are pushed through pre-push concurrently).
 2. Apply Patch A (`firebase-distribute.sh`) + update `docs/scripts/firebase-distribute.sh.md`.
 3. Apply Patch B (`.github/pre-push`).
 4. Re-run `bash tests/cycle-coverage/test_wiring.sh` (must stay 5/5) and `bash tests/cycle-coverage/test_cycle_coverage.sh` (must stay 7/7).
 5. Commit with the §3.4-style Bluff-Audit: stamp naming the gate-shaping files.
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5. What this does NOT change

- `scripts/check-cycle-coverage.sh` — untouched (already committed + tested).
 - `tests/cycle-coverage/test_cycle_coverage.sh` — untouched.
 - The `$6.P` / `$6.AA` / `$6.Z` gates already in `firebase-distribute.sh` — untouched; Gate 7 is additive and runs after them.
 - Checks 1–9 in pre-push — untouched; Check 10 is additive.
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6. Proof the wiring logic is sound (hermetic, no real files edited)

`tests/cycle-coverage/test_wiring.sh` generates STUB wrappers whose grafted blocks are byte-identical to `$1.2` and `$2.2`, invokes the REAL `scripts/check-cycle-coverage.sh` against synthetic `$6.Z` evidence + map fixtures (`HEAD` + now injected via `LAVA_CYCLE_COVERAGE_HEAD` / `LAVA_CYCLE_COVERAGE_NOW_EPOCH`), and asserts:

```
test: $6.AK GATE-WIRING contract (firebase-distribute Phase-1 Gate
+ pre-push Check 10)
  PASS [firebase_coverage_pass] exit=0 (expected 0) + matched
'$6.AK gate PASS'
  PASS [firebase_coverage_fail] exit=1 (expected 1) + matched
'FATAL $6.AK'
  PASS [prepush_advance_pass] exit=0 (expected 0) + matched
'Check 10 OK'
  PASS [prepush_advance_fail] exit=1 (expected 1) + matched
'$6.AK violation'
  PASS [prepush_no_advance] exit=0 (expected 0) + matched 'Check
10 OK'
-----
cases passed: 5    cases failed: 0
RESULT: PASS – firebase Phase-1 Gate + pre-push Check 10 wiring
proven against the real gate
```

Mutation rehearsal (swallow the refusal): the two negative cases flip to FAIL and the firebase stub prints "distribute would proceed" after a failing gate — the exact `$6.AK` incident shape the wiring exists to prevent. The wiring is therefore falsifiable and sound.